

32 Hydroxy compounds

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Phenol is more acidic than aliphatic alcohol because the conjugate phenoxide ion delocalises negative charge. Phenol activates its ring toward bromination and nitration. Comparing reactions with Na, NaOH and Na_2CO_3 distinguishes acidity levels: typical phenol reacts with NaOH but not sodium carbonate under usual conditions.

Worked example: follow the reasoning

Phenol + NaOH → sodium phenoxide + water, whereas ethanol does not react appreciably with aqueous NaOH.

Quick check · multiple choice

A Which reagent reacts with phenol but typically not ethanol in acid–base test? [1]

- A Water
- B Sodium metal
- C Bromine water only
- D Aqueous NaOH

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Phenol and ethanol both react with Na metal. Give one reagent that distinguishes them and expected result. [3]

Answers and explanation

Quick check: D. Aqueous NaOH

Question 1 [3 marks]

Aqueous NaOH [1]; phenol dissolves/reacts forming phenoxide [1]; ethanol shows no comparable acid-base reaction [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

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